

REMARKABLY PRECISE • REGULATORY OPERATIONS

The FDA Pre-Submission (Q-Sub) Playbook

A complete, no-experience-required, step-by-step guide to getting FDA's written read on ARKA as Non-Device Clinical Decision Support — done as cheaply as possible.

Free FDA program

Feedback, not approval

~70-day clock

Built for ARKA-CLIN + ARKA-INS

What this is: Your operating manual for filing a Pre-Submission with FDA's Digital Health Center of Excellence — what to fix in your existing brief, the new documents you must create, the cheapest viable path, the tricks that make reviewers comfortable calling you "Non-Device," and the click-by-click submission process.

What this is not: Legal or regulatory advice. It is a founder's field guide built from FDA's own current guidance and your codebase. Have a human with regulatory experience review the package before you file.

Prepared for **Arri Kanna, Founder — ARKA Health, Inc.** | June 2026 | Version 1.0

Based on the FDA CDS Software Guidance (Jan 6, 2026), the Q-Submission Program Guidance (May 29, 2025), and your current repository.

What's inside

Part 0 — Start here (the 60-second orientation)	3
Part 1 — What a Pre-Sub actually is, in plain English	5
Part 2 — ARKA's winning strategy: split the product	7
2.2 The five software exclusions & where each ARKA piece lands	7
2.4 Why ARKA-INS is "administrative," not CDS	7
Part 3 — Fix the brief first (make it match today's product)	10
Part 4 — The new documents you must create	14
4.0 Master document checklist	14
4.1 The Pre-Sub package 4.3 The gap-fillers	14
Part 5 — Doing it cheap: where \$0 works and where to spend	19
Part 6 — Tips & tricks to get labeled Non-Device CDS	21
Part 7 — The submission process, click by click	24
Part 8 — Timeline & what to run in parallel	27
Part 9 — Red flags that flip you to "device"	28
Appendices — Checklist, criteria map, glossary, links, email templates	29

PART 0

Start here

If you read only this page before diving in, you'll still know the shape of the whole project.

THE FIVE-SENTENCE VERSION

(1) A **Pre-Submission** is a free, voluntary request for FDA's written opinion on a regulatory question — here, "do you agree ARKA is not a medical device?" (2) FDA gives you **feedback, not approval**; you may never call the result a clearance. (3) Your strongest, cheapest move is to **split your product**: ARKA-CLIN is the Non-Device CDS question; ARKA-INS (coverage, prior-auth, cost) is "administrative support" software, a different and simpler exclusion; the DICOM thumbnail viewer is a separate function you keep out of scope. (4) Before you file, you must **fix the brief** (it still describes the old imaging-only "AIIE") and **create about a dozen new documents** — most of which you can write yourself from materials already in your repo. (5) The only money you should plan to spend is a few hours of a **licensed clinician's time** to sign off on your clinical content, plus an optional flat-fee review of your two highest-stakes paragraphs.

How to use this guide

Work through it in order once, then treat Parts 4 and 7 as your working checklists. Part 3 (fix the brief) and Part 4 (create the new documents) are the bulk of your effort — roughly one to three weeks of part-time work. Part 7 (the mechanics of submitting) takes a few days. The waiting is on FDA's side, about 70 days, which is exactly why you can start now and let it run in the background while you raise money and sell.

Throughout, you'll see colored boxes:

Box	Meaning
DO THIS	A concrete action to take.
WATCH OUT	A common mistake or trap, especially for first-timers.
NEVER	A line you must not cross — it can flip you to "device" or sink your credibility.
MONEY	A cost decision: what's free, what's worth paying for, and roughly how much.
TIP	An edge that makes reviewers comfortable saying "yes, Non-Device."
WHY FDA CARES	The reasoning behind a requirement, so you can satisfy its intent, not just its letter.

THE ONE MINDSET SHIFT THAT MATTERS MOST

Your current brief is written like a *verdict*: "AIE **is** not a device; this requires an Act of Congress to change." That tone is perfect for investors and dangerous for FDA. In a Pre-Sub you are a guest asking a question, not a lawyer declaring a holding. Reframe every assertion as a request for concurrence — "We have designed ARKA-CLIN to meet all four criteria; **does the Agency agree?**" Reviewers reward humility, specificity, and organization. They bristle at being lectured about the statute. This single change in posture runs through the entire package.

What a Pre-Sub actually is

Plain-English definitions first, so the rest of the guide reads easily.

1.1 The program, in one paragraph

The "free FDA checking place" is the **Q-Submission Program**, run by FDA's Center for Devices and Radiological Health (CDRH) and supported by its **Digital Health Center of Excellence (DHCoE)**. The most common Q-Sub is a **Pre-Submission** ("Pre-Sub"): you send FDA a written description of your software plus a short list of specific questions, and FDA sends back **written answers** and (if you ask) holds a meeting to discuss them. It is governed by the guidance *"Requests for Feedback and Meetings for Medical Device Submissions: The Q-Submission Program,"* which FDA finalized in its current form on **May 29, 2025**.

Question you have	The answer
Does it cost anything?	No. There is no user fee for a Pre-Submission. (Contrast: a "513(g) Request for Information," another way to ask FDA about classification, <i>does</i> carry a fee — see Part 5.)
How long does it take?	FDA's target is written feedback within ~70 days (it commits to meeting this for 90% of Pre-Subs). If you requested a meeting, it's scheduled around the same time. Add a ~15-day acceptance/technical screen at the front.
Is it voluntary?	Yes. Nothing forces you to do this. You're doing it to <i>de-risk</i> your "we're not a device" position and to have something credible to show investors and partners.
What do I get?	FDA's written view on your specific questions, plus optional meeting minutes. That's it — and that's valuable.
Who reads it?	A lead reviewer (and colleagues) at CDRH, often with DHCoE involvement for digital-health/CDS questions.

1.2 What it gives you — and what it absolutely does not

NEVER DESCRIBE A PRE-SUB RESULT AS APPROVAL

A Pre-Sub produces **feedback, not a clearance, approval, registration, or endorsement**. If FDA agrees you're Non-Device, the correct phrasing is: *"FDA provided written feedback concurring with our Non-Device CDS determination for ARKA-CLIN."* You may **never** say "FDA-approved," "FDA-cleared," "FDA-registered," or imply endorsement — in investor decks, on the website, on cards, anywhere. Misusing FDA's name is its own violation and will undo all your careful work. Your repo already scans clean for these words; keep it that way.

1.3 Why a one-person company can do this solo

The Pre-Sub program was built for exactly this: early, low-stakes, written dialogue. The package is short relative to a real device submission, the template is free and self-guiding, and the channel (the CDRH Portal) is open to anyone who registers. Thousands of small sponsors file Pre-Subs without large regulatory teams. What you cannot fully DIY is *clinical credibility* — FDA will expect that a licensed clinician stands behind your clinical logic. That's the one place to spend a little (Part 5).

PLAIN-ENGLISH GLOSSARY (THE EIGHT WORDS YOU'LL KEEP SEEING)

Device — legally, software FDA regulates. Your whole goal is to confirm ARKA's functions are *not* devices.

Non-Device CDS — clinical decision support that meets four criteria in the law and is therefore excluded from "device." **520(o)(1)** — the section of the law listing software that isn't a device; paragraph (E) is CDS, paragraph (A) is administrative support. **Intended use / indications** — what you say your software is for; the most important words you'll write. **Enforcement discretion** — when something *is* technically a device but FDA chooses not to enforce; weaker than exclusion. **CDRH Portal** — the website where you upload the submission. **PreSTAR** — the free fill-in-the-blanks template that becomes your submission. **DHCoE** — FDA's Digital Health Center of Excellence, your friendly first point of contact.

1.4 Where a Pre-Sub fits among your free FDA options

You actually have three escalating ways to talk to FDA, and you should use them in order — cheapest and most informal first:

Channel	Cost	Best for	When for ARKA
Digital Health Policy Navigator (self-service web tool)	Free	Walking your own product through FDA's 7-step logic to predict the answer yourself.	Now. Do this before anything else; it sharpens your questions.
Digital Health mailbox (email DHCoE)	Free	An informal gut-check question, no formal record, fast.	Before filing — sanity-check your framing with a short note.
Pre-Submission (Q-Sub)	Free	A formal, written, citable answer to specific questions, plus a meeting.	The main event — what this guide builds toward.

Your existing brief already lists this exact sequence in its "Planned FDA Engagement" section. Good instinct — this guide just makes it concrete.

ARKA's winning strategy

The single highest-leverage decision in this whole project: stop treating ARKA as one thing. Split it into functions, and give each function the cleanest exclusion the law offers.

2.1 Your product grew; your brief didn't

Your February 2026 brief describes one product — the "ARKA Imaging Intelligence Engine (AIIE)" — and argues it's Non-Device CDS. But your repository today is a unified platform with at least three distinct software functions:

Function	What it does (from your code)	Lives in
ARKA-CLIN	Imaging-appropriateness recommendations at order-select / order-sign over CDS Hooks; rules-first, ML-as-refinement, SHAP rationale + citations.	app/api/cds-services/ arka-clin-*, lib/cds- platform/*, ml-service/
ARKA-INS	Payer coverage intelligence, prior-authorization documentation (Da Vinci CRD/DTR/PAS), gold-carding, out-of-pocket cost estimates, shoppable-site comparison, appointment-booking checks.	app/api/ins/*, app/api/cds-services/ arka-ins-*, lib/ davinci/*, lib/aie/ oop-*
Reference viewer	Renders previously-acquired DICOM studies as non-diagnostic thumbnails so a user can identify a prior study. Already scoped out of CDS in docs/ SCOPE_BOUNDARY.md .	lib/viewer/*, app/api/ ins/viewer/*

Your own `REGULATORY_RATIONALE_MEMO.md` already uses the modern names (ARKA-CLIN, ARKA-INS) and already commits to "rules-first" architecture. The brief is simply a version behind. The fix isn't to bolt INS onto the CDS argument — it's to recognize that **INS isn't CDS at all, and that's good news**.

2.2 The five software exclusions — and where each ARKA piece lands

Congress (via the 21st Century Cures Act) wrote **five** categories of software that are *not* devices, in section 520(o)(1) of the Food, Drug & Cosmetic Act. Everyone fixates on (E), CDS. But (A) is the unsung hero for half of your product:

¶	Excluded software category	Plain meaning	ARKA piece
(A)	Administrative support of a health care facility	Claims/billing, determination of health-benefit eligibility , appointment scheduling, business analytics, population health, and analysis of historical claims data to predict future utilization or cost-effectiveness .	ARKA-INS coverage, gold-card, OOP cost, scheduling
(B)		Wellness apps unrelated to diagnosis/treatment.	—

	Maintaining/encouraging a healthy lifestyle		
(C)	Electronic patient records	EHR-type record keeping.	—
(D)	Transfer / store / convert / display data	Plumbing that moves or shows data/results.	(arguably the viewer's display)
(E)	Clinical Decision Support (four criteria)	Recommendations to a clinician they can independently review.	ARKA-CLIN

WHY THIS IS THE MOST IMPORTANT TABLE IN THE GUIDE

The four CDS criteria are a *narrow, demanding* test. Trying to force coverage, cost, and prior-auth through them is hard and unconvincing — a coverage card isn't "a recommendation about prevention, diagnosis, or treatment of a disease," and a denial-risk flag arguably leans on the clinician rather than inviting independent clinical review. But you don't have to. Paragraph (A) was written for *exactly* this software: eligibility determination and claims-based cost prediction. Routing INS to (A) means it never has to pass the CDS criteria at all. You convert your biggest regulatory liability into a footnote.

2.3 ARKA-CLIN → Non-Device CDS under (E)

This is the genuine CDS question and the heart of your Pre-Sub. ARKA-CLIN is a strong (E) candidate because, by design, it: takes only structured data (no image pixels/signals); draws on published guidelines and peer-reviewed literature; presents ranked options with scores, not a single directive; and exposes the basis (citations + SHAP factors) so a clinician can independently review it. Part 6 walks each criterion in detail. Your job in the Pre-Sub is to ask FDA to **concur** on CLIN specifically.

2.4 ARKA-INS → Administrative support under (A)

Map each INS feature to the (A) language and the fit is obvious:

ARKA-INS feature	Statutory (A) hook
Coverage / eligibility cards	"determination of health-benefit eligibility"
Gold-card auto-approval scoring	"analysis of historical claims data to predict future utilization or cost-effectiveness"
Out-of-pocket estimate, GFE, shoppable-site comparison	"processing and maintenance of financial records"; price transparency
Prior-auth documentation (Da Vinci CRD/DTR/PAS)	claims/administrative workflow; FDA has signaled documentation tools are administrative support, not CDS
Appointment-booking / site optimization	"appointment schedules," practice management

TIP — SAY IT PLAINLY, THEN ASK

In the package you'll *state* ARKA-INS is administrative-support software under 520(o)(1)(A) and pose it as a confirmation question, not an argument: *"ARKA-INS performs benefit-eligibility determination and claims-based utilization analysis; we treat these as administrative-support functions under §520(o)(1)(A). Does FDA concur these functions are outside device authority?"* Short, confident, answerable.

2.5 The reference viewer → a separate, out-of-scope function

Your baseline audit honestly flags that `lib/viewer/*` actually parses DICOM and extracts JPEG pixels for thumbnails. That's real image handling — and Criterion 1 of CDS forbids image processing. The good news: it's a *different function* from CLIN, it doesn't feed any recommendation, and your `SCOPE_BOUNDARY.md` + import-lint already wall it off. You have two clean options, and you should pick one before filing:

- **Option A (cleanest): carve it out of the product you're asking about.** Tell FDA your Pre-Sub covers ARKA-CLIN and ARKA-INS; the reference viewer is a separate display utility not part of the functions under review, and never contributes to a recommendation.
- **Option B: argue the viewer itself is (D) display.** Defensible, but it invites questions you don't need right now. Save it.

WATCH OUT — MAKE THE FIREWALL REAL, NOT JUST DOCUMENTED

Your memo says a CI guard (`scripts/regulatory-checks.ts`) will fail any build that imports image/signal code into CLIN paths — but it's marked *TODO, not yet wired*. Before you cite this protection to FDA, actually turn it on. A control you describe but don't run is the kind of gap a sharp reviewer finds. Wiring it is an afternoon of work and becomes strong evidence (see Part 4).

2.6 The "multiple-function" framework — your legal cover for being one app

FDA expects products to have several functions. Its *"Multiple Function Device Products: Policy and Considerations"* guidance says FDA reviews the specific "function-under-review" and considers how other functions affect it. Translated for you: you don't have to pretend ARKA is only CDS. You present a clean **function map** (CLIN = (E); INS = (A); viewer = out of scope), and ask FDA to confirm each. This is both honest and strategically strong — it shows you understand the framework better than most first-time filers.

DO THIS — WRITE YOUR HEADLINE ASK NOW

One paragraph you'll refine and reuse everywhere: *"ARKA is a single platform with distinct software functions. ARKA-CLIN provides imaging-appropriateness clinical decision support that we have designed to meet all four Non-Device CDS criteria under §520(o)(1)(E). ARKA-INS provides benefit-eligibility and claims-based administrative functions we treat as excluded under §520(o)(1)(A). A reference-image viewer that displays prior studies is a separate function, outside the scope of this submission. We request FDA's concurrence with these determinations."* Everything else in the package supports those four sentences.

Fix the brief first

Your brief is a strong foundation, but it describes a product you've outgrown and takes a tone built for investors, not regulators. Here is every gap I found between the brief and your current codebase, and exactly how to fix each one. Treat this as a punch list.

3.0 The punch list at a glance

#	Issue in the current brief	Why it's a problem for FDA	Severity
1	Calls the product "AIIE / ARKA Imaging Intelligence Engine" only	Doesn't match your product (ARKA-CLIN + ARKA-INS). FDA needs the function under review named consistently.	High
2	Lumps INS coverage/PA/cost into the CDS argument	Coverage/cost isn't CDS; forcing it weakens the whole package. Route to (A).	High
3	Intended use says "adults 18+ only"	Your model card and rules include <i>pediatric</i> cases (e.g., pediatric RLQ pain). Self-contradiction on the single most important sentence.	High
4	Cites guidance "GUI01400062"	That document number looks invented. The Jan 6, 2026 guidance is <i>real</i> , but cite it correctly or a reviewer doubts everything.	Medium
5	"AIIE never blocks orders; override requires no justification"	INS code uses "critical" cards, a <code>buildOrderSignCriticalBlockCard</code> , "override required to proceed," and a "Cancel order" label. Reviewer will see the contradiction.	High
6	Compliance checklist marks validation "✓ Complete"	Your model is trained on <i>synthetic</i> data (RMSE ≈ 1.41 , $\sim 74\%$ 3-class). Calling it complete overstates; honesty is safer and expected.	High
7	Built on RAND/UCLA + ACR sources	Your memo's evidence policy says "never ACR," yet citations include ACR items. Pick one story; FDA's 2026 guidance stresses "well-understood and accepted sources."	Medium
8	Tone: "statutory exclusion... requires Congressional action to change"	Reads as lecturing FDA on the law. Reframe as a request for concurrence.	Medium
9	Declares all four criteria "MET" as conclusions	In a Pre-Sub you ask, not declare. Keep the analysis; change the verbs.	Medium
10	Contact is <code>arrihantk@gmail.com</code> ; phone listed	Use one consistent professional contact across every document and the Portal account.	Low

11	References evidence files that don't all exist	DOC-VAL-001, the Loom URL, the Phase-12 SHAP screenshot, a sandbox HTML report are cited but missing from the repo. Don't cite what you can't attach.	Medium
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3.1 Rename consistently (Issue 1)

Decide your public nomenclature and use it everywhere — brief, package, cards, website. Recommended: **"ARKA-CLIN"** (the CDS function) and **"ARKA-INS"** (the administrative function), with "AIIE" demoted to what it actually is in your code — the internal scoring engine, not the product. Do a find-and-replace pass; mismatched names across documents are exactly what makes a reviewer think the package was assembled carelessly.

3.2 The guidance citation (Issue 4) — good news, with a fix

YOU WERE RIGHT THAT THERE'S A 2026 GUIDANCE — JUST CITE IT PROPERLY

FDA **did** issue a revised CDS Software guidance on **January 6, 2026** (re-issued January 29, 2026), superseding the September 2022 version. So your instinct is correct and current. But drop the invented-looking "GUI01400062." Cite it as: "FDA, Clinical Decision Support Software — *Guidance for Industry and FDA Staff, issued January 6, 2026 (re-issued January 29, 2026), available at [fda.gov/media/109618/download](https://www.fda.gov/media/109618/download).*" Citing it cleanly signals you've actually read the current guidance — which most filers haven't yet.

TIP — TURN THE 2026 CHANGES INTO A BRAG

The 2026 guidance newly extends *enforcement discretion* to single-output CDS (where only one recommendation is clinically appropriate). Your brief proudly notes ARKA presents *multiple* ranked options and therefore "does not rely on enforcement discretion." Keep that — it's a sophisticated point that puts you in the stronger *statutory exclusion* lane, not the weaker discretion lane. It tells the reviewer you know the difference.

3.3 Reconcile the intended-use / population mismatch (Issue 3)

This is your most important sentence, and right now it disagrees with itself across documents. Your brief says adults only; your model card says "adult and pediatric outpatient"; your rules include pediatric appendicitis. Before filing, **decide**: either (a) restrict the intended use to adults and remove pediatric rules/citations from the in-scope product, or (b) include pediatrics and own it in the intended use, model card, and validation plan. Whichever you choose, make all documents agree. (Part 4 gives you a drafted intended-use statement; Part 6 gives the drafting rules.)

3.4 Stop shoeorning INS into CDS (Issue 2) & reconcile the "never blocks" claim (Issue 5)

These two are the same root problem and the same fix. Once ARKA-INS is reframed as **administrative support under (A)**, its payer-driven behaviors stop being a CDS problem:

- The "critical" cards and "override/DTR required to proceed" language in `lib/davinci/crd.ts` are **payer-adjudication semantics** (Da Vinci CRD), not clinical coercion. As administrative-support software, INS isn't held to CDS Criterion 3 at all.
- For **ARKA-CLIN** specifically, keep the "recommendations, not directives; override needs no justification" promise — and make sure no CLIN card uses blocking/critical framing or a "Cancel order" suggestion. Audit `lib/cards/*` and the CLIN converter so the non-coercion claim is literally true for the CDS function.

WATCH OUT — YOUR "CANCEL ORDER" LABEL AND CRITICAL-TIER CARDS

Your baseline audit found a `label:"Cancel order"` suggestion and several `indicator:"critical"` cards with mandatory-override framing. If any of these are on a *CLIN* card, soften them before you tell FDA CLIN never coerces. If they're only on *INS* cards, that's acceptable under (A), but say so explicitly so the reviewer doesn't assume it's CLIN.

3.5 Be honest about validation (Issue 6) & sources (Issue 7)

Your model is trained on **synthetic, guideline-aligned data** — that's fine for a Pre-Sub, and pretending otherwise is the real risk. Change the checklist from "Validation Study Results ✓ Complete" to something like *"Synthetic validation complete; real-world retrospective validation planned"* and make "is our synthetic evidence sufficient for the stated intended use?" one of your FDA questions (it's a great question — see Part 4). On sources, pick a lane: if you're avoiding ACR for licensing reasons, remove ACR citations from cards and lean on NICE/USPSTF/ACR-free guidelines consistently; if you're using ACR, drop the "never ACR" language from the memo. The 2026 guidance specifically rewards recommendations grounded in "well-understood and accepted sources," so make your source story airtight and consistent.

3.6 Shift the tone from verdict to request (Issues 8 & 9)

Brief says (verdict tone)	Change to (Pre-Sub tone)
"AIIE meets all four statutory criteria... No 510(k)... is required."	"We have designed ARKA-CLIN to meet all four criteria; we request FDA's concurrence."
"This classification requires Congressional action to change."	(Delete. True, but it reads as lecturing the Agency.)
"✓ MEETS CRITERION" (×4)	"How we meet Criterion X" + "Question: does FDA agree?"
"the strongest available regulatory pathway"	(Delete the salesmanship; let the analysis speak.)

3.7 Clean up references and contact (Issues 10 & 11)

Use one professional email and one phone across the Portal account, cover letter, and every document. And before you cite an exhibit, confirm it exists and you can attach it. Your memo references a Loom URL file, a Phase-12 SHAP screenshot, and a sandbox HTML report that aren't in the repo. Either generate them (Part 4) or remove the reference. **Citing evidence you can't produce is worse than not citing it.**

NET RESULT OF PART 3

After this pass you'll have a brief that (a) names the real product, (b) routes INS to (A), (c) tells one consistent story about scope, population, sources, and validation, and (d) asks rather than declares. That cleaned-up brief becomes the backbone of your Pre-Sub package in Part 4 — much of it can be reused almost verbatim.

PART 4

The new documents you must create

This is the part you asked for most. Below is the complete master checklist, then a spec for every document — what it is, why FDA cares, what "great" looks like, where your existing material lives, and roughly the effort and cost. I've also drafted the core package documents for you (delivered alongside this guide in [docs/regulatory/q-sub-draft/](#)).

4.0 Master document checklist

Three statuses: **Have** you already have it in some form; **Fix** exists but needs updating per Part 3; **Create** brand new. "Drafted" means I've written a first draft for you.

#	Document	Status	Where your material is	Effort
A. The Pre-Sub package (what you upload)				
1	Cover letter	Create · Drafted	New (template provided)	1–2 h
2	Submission cover sheet / admin info	Fix	PreSTAR captures most of it	1 h
3	Product (function) description	Fix · Drafted	Brief §2; INS_ARCHITECTURE.md	3–5 h
4	Intended-use / indications statement	Create · Drafted	Brief §2.2 (needs reconciling)	2–4 h
5	The 3–4 FDA questions	Create · Drafted	Memo §8	2 h
6	Multiple-function analysis (CLIN/INS/viewer map)	Create · Drafted	New; uses SCOPE_BOUNDARY.md	2–3 h
7	Meeting request (type, format, 3+ dates, agenda, attendees)	Create · Drafted	New (template provided)	1 h
B. Supporting evidence you attach				
8	Regulatory rationale / four-criteria analysis (revised brief)	Fix	The PDF brief + REGULATORY_RATIONALE_MEMO.md	4–6 h
9	Model card	Have	ml-service/MODEL_CARD.md	1 h polish
10	Data-input inventory + data-flow diagram	Have	Brief §3.1 / §5.3	1 h
11	Sample screens / card mock-ups (transparency & usability)	Have	docs/regulatory-evidence/sandbox-screenshots/	1–2 h
C. The gap-fillers (new substance FDA will look for)				
12		Create		clinician hrs

	Clinical sign-off (≥1 licensed clinician entry)		CLINICAL_SIGN_OFF_LOG.md (0 signed)	
13	ARKA-INS administrative-support memo (520(o)(1)(A))	Create · Drafted	New	2–3 h
14	Usability / human-factors note (Criterion 4)	Create	Brief checklist marks this "Planned"	3–5 h
15	Validation summary, written honestly (synthetic now)	Fix	MODEL_CARD + metrics.json	2–3 h
16	Change-control plan (rules, catalogue, model weights)	Fix	Memo §7 (skeleton exists)	2–4 h
17	Criterion-1 firewall evidence (wire the CI guard)	Create	scripts/regulatory-checks.ts (TODO)	0.5–1 day
18	Labeling / no-misbranding attestation	Create · Drafted	fda-disclosure.ts; banner copy	1 h

MONEY — THE ONLY LINE ITEM WITH A REAL DOLLAR COST

Everything above is DIY **except #12, the clinician sign-off**. That requires a licensed clinician's hours. Budget for it; see Part 5 for how to get it cheaply (often free via an advisor). Optionally, a flat-fee expert review of #4 (intended use) and #5 (questions) is worth considering — those two artifacts carry the most lasting consequence.

4.1 The Pre-Sub package documents (detailed specs)

① Cover letter — Drafted for you

What it is: A one-page letter that says who you are, that this is a Pre-Submission under the Q-Sub program, a two-sentence product summary, and a one-line statement of what you're asking. **Why FDA cares:** It's the reviewer's first impression and routing aid. **Great looks like:** Crisp, confident-but-humble, names the function(s) under review, states "we request written feedback and a teleconference," lists your 3–4 questions by title. Keep it to one page.

② Submission cover sheet / administrative info

What it is: Sponsor name, contact, submission type (Pre-Sub), a proposed Q-Sub "purpose." **Where:** PreSTAR collects nearly all of this through its form fields, so you mostly fill blanks rather than write prose. Just make sure the contact matches your Portal account.

③ Product (function) description — Drafted for you

What it is: A few pages describing what ARKA-CLIN and ARKA-INS do, how the CDS Hooks integration works, what data flows in and out, and — crucially — that the clinician sees the rationale and citations and remains the decision-maker. **Why FDA cares:** This is where the Non-Device story is shown, not asserted. **Great looks like:** includes your three-layer data-flow diagram, the "rules-first, ML-as-refinement" architecture, an annotated

screenshot of a CLIN card showing the citation above the score, and a clean statement of what the software deliberately does *not* do (no image analysis, no autonomous ordering, no time-critical alerts).

④ **Intended-use / indications statement** — Drafted for you

THIS IS THE MOST CONSEQUENTIAL PARAGRAPH YOU WILL EVER WRITE FOR ARKA

FDA's entire determination keys off intended use. Every word is load-bearing. Get it reviewed by a human. Part 6.1 gives the drafting rules; the draft in your `q-sub-draft` folder is a strong starting point, deliberately written to land inside the four criteria.

⑤ **Your 3–4 questions** — Drafted for you

What it is: The specific things you want FDA to answer. **Why it matters:** FDA recommends **no more than four primary topics** per Pre-Sub, and fewer, sharper questions get better answers. **Great looks like:** closed, concurrence-style questions tied to your analysis. Your drafted set (refined from your memo §8):

1. *Does FDA concur that ARKA-CLIN, as described, meets the four Non-Device CDS criteria under §520(o)(1)(E)?*
2. *Does FDA concur that ARKA-INS's benefit-eligibility and claims-based functions are administrative-support functions under §520(o)(1)(A), outside device authority?*
3. *For ARKA-CLIN to maintain Non-Device CDS status, what does FDA expect for change-control over the rule library, feature catalogue, and model weights?*
4. *Is our synthetic-validation and sandbox evidence package sufficient to support the stated intended use, or does FDA expect prospective/retrospective clinical data?*

⑥ **Multiple-function analysis** — Drafted for you

What it is: A short table/narrative mapping each function to its regulatory basis (CLIN → (E); INS → (A); viewer → out of scope) per FDA's multiple-function guidance. **Why FDA cares:** It proves you understand that a multi-feature app can still be non-device, and it pre-empts the "but what about the viewer / the coverage cards?" questions.

⑦ **Meeting request details** — Drafted for you

FDA requires, for the meeting you're requesting: **meeting type** (Pre-Sub), **requested format** (teleconference is normal and free), **three or more preferred dates/times** (spread them 70–90 days out), a **draft agenda with minutes allotted per item**, and a **list of attendees with titles/affiliations** — and you must **flag any non-U.S. citizens** for FDA's logistics. Keep your side small: you, and ideally your clinical advisor.

4.3 The gap-fillers (where the real new work is)

⑫ Clinical sign-off — the one that needs a human (highest priority)

WHY FDA CARES — AND WHY YOUR OWN REPO FLAGS THIS

Criterion 2 expects recommendations grounded in accepted clinical sources, and Criterion 4 expects a basis a clinician can trust. Your `CLINICAL_SIGN_OFF_LOG.md` says it plainly: it is "required reading for any FDA Q-Submission," and without "at least one signed entry... ARKA cannot defensibly claim that its rule library and feature catalogue are clinically curated." Right now it has **zero** signed entries. This is the single biggest substantive gap.

What to do: Get one licensed clinician (MD/DO/NP/PA, active license) to review your rule library, citations, feature catalogue, and card language, and sign at least one dated entry with name, credentials, and license number. **Great looks like:** a signed entry per open TODO in the log, plus a one-paragraph reviewer statement you can attach. See Part 5 for getting this cheaply.

⑬ ARKA-INS administrative-support memo — Drafted for you

What it is: A 1–2 page memo asserting and supporting that INS functions fall under §520(o)(1)(A), feature-by-feature (the table from Part 2.4). **Why:** It's the substance behind your Question 2 and keeps INS out of the CDS analysis entirely.

⑭ Usability / human-factors note (Criterion 4, 2026 emphasis)

TIP — THE 2026 GUIDANCE RAISED THE BAR HERE, SO LEAN IN

The January 2026 guidance puts **much stronger emphasis on usability**: information should "prioritize decision-relevant details and avoid information overload." Your three-tier disclosure model (score+top factors → expanded SHAP → full methodology) is a perfect answer — if you frame it as a usability design choice. Write a short note explaining how a busy clinician sees the citation and top factors immediately, with depth one click away. Your brief lists this as "Planned — pending clinical advisor input"; finishing it is now a differentiator, not a checkbox.

⑮ Honest validation summary

One page: what the model is (XGBoost), what it was trained on (5,000 synthetic, guideline-aligned examples), the held-out metrics (RMSE $\approx$ 1.41; $\sim$ 74% 3-class accuracy), the explicit statement that this is *not* a production clinical-performance claim, and your plan for real-world retrospective validation. Pair it with your Question 4. Honesty here is a strength: you're asking FDA what evidence they'd want, not pretending you already have it.

⑯ Change-control plan

What it is: How you version and gate changes to the rule library, citation registry, feature catalogue, and model weights without drifting out of Non-Device status. **Where:** Your memo §7 already sketches this (clinician sign-off required for rule/catalogue changes; CI enforcers for card basis and Criterion-1 import boundaries). Turn the sketch into a standalone 2-page SOP. **Why FDA cares:** It's literally your Question 3 — show them you've thought about it.

⑪ Criterion-1 firewall evidence — wire the guard

What it is: Proof that no image/signal code can enter the CLIN path. **Do:** finish `scripts/regulatory-checks.ts` (currently TODO) so CI fails on any DICOM/waveform import into in-scope CDS code, and run it. Attach a passing CI screenshot. Combined with `SCOPE_BOUNDARY.md` and the import-lint, this turns a claim into demonstrable engineering.

⑫ Labeling / no-misbranding attestation — Drafted for you

What it is: A short statement (and an internal grep result) showing your marketing never says "FDA-approved/cleared/registered/endorsed," and that cards carry the correct Non-Device disclosure. Your baseline audit shows zero forbidden hits today — capture that as evidence and keep a CI check so it stays true.

DO THIS — ASSEMBLE IN THIS ORDER

(1) Fix the brief (Part 3). (2) Write the four core package docs — already drafted for you. (3) Get the clinician sign-off in motion (longest lead time — start week 1). (4) Wire the CI guard. (5) Polish the supporting evidence. (6) Drop everything into PreSTAR (Part 7).

Doing it cheap

You asked to do this as cheaply as possible and only pay where absolutely necessary. Here's the honest cost map: almost all of it is genuinely free, and there's exactly one thing I'd insist you spend on.

5.1 The free path covers ~90% of this

No part of the Pre-Sub itself costs money. The template is free, the Portal is free, the submission has no user fee, and the meeting is a free teleconference. Every document in Part 4 except the clinician sign-off can be written by you (with the drafts I've provided as a head start). The free FDA channels below are underused by founders:

Free resource	What it does for you
Digital Health Policy Navigator	FDA's own 7-step web tool to self-classify each software function. Run CLIN and INS through it; screenshot your answers as supporting evidence.
Digital Health mailbox (DigitalHealth@fda.hhs.gov)	Email DHCoE an informal framing question before you file. Free, no formal record, can shape your questions.
DICE — Division of Industry & Consumer Education	CDRH's help desk for "how do I..." submission mechanics. Call/email them when stuck on the Portal or PreSTAR.
The guidances themselves	The CDS guidance (Jan 2026), the Q-Sub guidance (May 2025), and the Multiple-Function guidance are free PDFs. Reading them beats paying someone to summarize them.
PreSTAR help text	The template has built-in instructions in every section. It is, in effect, a free regulatory consultant for formatting.

5.2 The two things worth paying for (and only these)

SPEND #1 (DO THIS): A LICENSED CLINICIAN'S SIGN-OFF — OFTEN \$0 IF YOU HAVE AN ADVISOR

Your package's credibility rests on a clinician standing behind your clinical content. Options, cheapest first: **(a)** a clinical advisor/co-founder who'll review for equity or goodwill — \$0; **(b)** a moonlighting physician for a few hours of focused review — a few hundred dollars; **(c)** a telehealth/"physician advisor" marketplace. You need *hours*, not a retainer. Give them your rule list, citations, and card language, and the sign-off log to date and initial.

SPEND #2 (OPTIONAL BUT SMART): A FLAT-FEE REVIEW OF YOUR TWO HIGHEST-STAKES PARAGRAPHS

If you'll pay for anything regulatory, make it a **one-time, flat-fee** review of just the *intended-use statement* and the *four questions* by a digital-health regulatory consultant. These two artifacts have the longest-lasting consequences, and a 1–2 hour expert read is high-leverage. Ask explicitly for a **fixed price for a defined scope** ("review and redline two pages") — not an open-ended engagement. This is the "willing to pay if absolutely necessary" money, used surgically.

5.3 Cost tiers at a glance

Tier	Rough cost	What you get	When it's right
Tier 0 — Pure DIY	\$0	You write everything; a friendly clinician signs off; you file solo.	If you have any clinician in your network. Recommended starting point.
Tier 1 — DIY + targeted help	~\$500–\$2k	Tier 0 plus a few paid clinician hours and/or a flat-fee review of intended-use + questions.	If you have no clinician handy, or want one expert read of the riskiest paragraphs.
Tier 2 — Consultant-assembled	~\$5k–\$15k+	A consultant assembles/reviews the whole Pre-Sub.	Only if you raise money and decide speed/peace-of-mind is worth it. Not necessary to start.

5.4 How to buy help without overpaying

- **Scope it as a fixed deliverable.** "Redline these two pages by Friday for \$X," not "advise us on regulatory strategy."
- **Prefer fractional/independent consultants** over big firms for a first Pre-Sub. Search "digital health / SaMD regulatory consultant Pre-Submission," and ask in founder communities. The professional society is **RAPS** (Regulatory Affairs Professionals Society) — a reasonable place to find independents.
- **Bring them a finished draft.** Reviewing your draft is far cheaper than authoring from scratch. The drafts in your `q-sub-draft` folder exist precisely to minimize paid hours.
- **Never pre-pay for a 510(k) or QMS.** If anyone's first suggestion is to start a 510(k) or build a Quality System, stop — that's the opposite of your strategy. You're confirming you're *not* a device.

WATCH OUT — THE ONE PLACE "CHEAP" CAN BACKFIRE

Skipping the clinician sign-off to save money is a false economy. It's the gap most likely to draw an FDA comment and the one thing you can't credibly fake. Spend the few hours (or call in a favor) here even if you DIY everything else.

Tips & tricks to get labeled Non-Device CDS

This is the "make the reviewer comfortable saying yes" section. Everything here is about framing your true, existing design in the language FDA's 2026 guidance rewards.

6.1 The intended-use sentence — drafting rules

Reviewers test your intended use against each criterion, word by word. Engineer it to pass:

Do include	Avoid
"for use by licensed health care professionals"	any hint of patient/consumer use (fails Criterion 3 framing)
"to support, not replace, the clinician's independent judgment"	"diagnoses," "detects," "determines" (sounds like a device output)
"based on structured clinical data" + "published guidelines and peer-reviewed literature"	"analyzes images/scans/signals/waveforms" (fails Criterion 1)
"provides ranked, evidence-cited options the clinician can independently review"	"alerts," "real-time," "time-critical," "automatically orders" (fails C3/C4)
"for elective and non-urgent imaging decisions"	"emergency," "triage," "stroke/sepsis/STEMI"

Drafted target (in your q-sub folder): "ARKA-CLIN is intended for use by licensed health care professionals to support the selection of clinically appropriate diagnostic imaging for adult [and pediatric — decide] patients in elective and non-urgent settings. Using structured clinical data, it presents ranked, evidence-cited imaging options drawn from published clinical guidelines and peer-reviewed literature, together with the basis for each option, so that the clinician can independently review the recommendation and retains full responsibility for the decision. ARKA-CLIN does not analyze medical images or physiological signals, does not provide time-critical alerts, and does not place orders."

6.2 The four criteria, mapped to ARKA and the 2026 changes

Crit.	What FDA wants (2026)	How ARKA-CLIN shows it
1 Inputs	No acquiring/processing/analyzing images, IVD signals, or signal-acquisition patterns.	Structured FHIR only; documented input inventory; <i>wired</i> CI guard blocking image/signal imports; viewer fenced off as a separate function.
2 Medical info	Display/analyze medical info from "well-understood and accepted sources" (guidelines, peer-reviewed literature).	Every card carries a <code>medicalBasis</code> with citation; consistent source policy; clinician-reviewed.
	Recommendations/options to an HCP, not a directive. 2026: single	

3 Recommendations	outputs now get <i>enforcement discretion</i> — but multi-option keeps you in stronger <i>statutory exclusion</i> .	Ranked options with 1–9 scores; non-blocking; override needs no justification; no "Cancel order" on CLIN cards.
4 Independent review	HCP can review the basis; not meant to rely primarily on the tool. 2026: stronger usability emphasis (no information overload); time-critical concern now lives <i>here</i> .	Three-tier transparency (immediate → expanded SHAP → methodology); citations one tap away; explicitly non-time-critical.

TIP — TURN THE 2026 "USABILITY" EMPHASIS INTO YOUR DIFFERENTIATOR

Most CDS filers treat transparency as a data dump. The 2026 guidance specifically warns against information overload. Your tiered design is the textbook answer — present it as a deliberate *usability* decision, with a sample screen showing the clinician gets the citation + top three factors at a glance and depth on demand. This is the easiest place to look more thoughtful than the average submission.

6.3 Ask questions that get clean answers

- **Closed beats open.** "Does FDA concur that..." yields a usable yes/clarify. "What does FDA think about our approach?" yields a non-answer.
- **Tie each question to your own analysis.** Pre-state your position, then ask for concurrence. Reviewers answer faster when you've done the work.
- **Respect the 4-topic limit.** FDA explicitly recommends ≤4 primary topics. Your four (Part 4.1 ⑤) are already at the ceiling — resist adding more.
- **Don't bury a hard question.** Lead with the CLIN concurrence question; it's your main event.

6.4 What makes a reviewer nervous — and how to defuse it

Reviewer worry	Your pre-emptive answer in the package
"There's a DICOM viewer in here — is this image analysis?"	Scope statement + import-lint + CI guard proving the viewer never feeds a recommendation.
"Coverage/denial cards sound like they pressure the clinician."	INS is administrative (A); CLIN is the only CDS function and is non-coercive by design.
"The model is a black box."	Rules-first; ML only refines; SHAP + citation for every factor; falls back to rules when ML is offline.
"Trained on synthetic data?"	Stated openly + a real-world validation plan + you're asking FDA what they'd want (Question 4).
"Who says the clinical logic is sound?"	Signed clinician entries in the sign-off log.

6.5 Language discipline

NEVER USE THESE WORDS ABOUT ARKA

"FDA-approved," "FDA-cleared," "FDA-registered," "FDA-certified," "FDA-endorsed." Also avoid clinical-device verbs for CLIN: "diagnoses," "detects disease," "alerts," "real-time monitoring." Preferred framing throughout: *"designed to meet the four Non-Device CDS criteria,"* and *"supports, not replaces, the clinician."*

6.6 Make it effortless to say yes

Organize the package so a busy reviewer can verify each criterion in minutes: a one-page function map up front, a one-page criteria → evidence table, consistent names, working cross-references, and every cited exhibit actually attached. Polished organization signals a sponsor who will be easy to work with — and reviewers extend more good faith to those sponsors.

The submission process, click by click

Once your package is written and reviewed, the mechanical part is a few days of focused work. Follow these stages in order.

A Stage A — Free pre-work (do before you touch the Portal)

1. **Run the Digital Health Policy Navigator** for ARKA-CLIN and ARKA-INS separately. It walks 7 steps of questions and points you to the relevant policies. Screenshot your path; it becomes supporting evidence and confirms your self-classification.
2. **Email the Digital Health mailbox** (DigitalHealth@fda.hhs.gov) a short, informal note describing the product and your proposed framing (CLIN=(E), INS=(A)). You may get a quick informal steer that sharpens your questions. (Template in Appendix F.)
3. **Finalize the package** from Part 4 and get your human review done (clinician sign-off + optional flat-fee read of intended use + questions).

B Stage B — Register for the CDRH Portal (1 day)

1. Go to FDA's page: "*Send and Track Medical Device Premarket Submissions Online: CDRH Portal.*" (Link in Appendix E.)
2. Create an account and verify your email. Anyone can register; you don't need an FDA establishment registration or a device listing to file a Pre-Sub.
3. Use the **same name, email, and organization** you'll put on the cover letter and in PreSTAR. Consistency matters for routing.

WATCH OUT — ACCOUNT APPROVAL CAN TAKE A LITTLE TIME

Don't leave registration to the last day. Set up the Portal account early (it can run in parallel with writing), so it's ready when your PDF is.

C Stage C — Download and open PreSTAR (the gotchas matter)

FDA's free template for Pre-Subs is the "**Early Submission Requests eSTAR (PreSTAR).**" As of the June 2026 update, the current version is **PreSTAR2 v3.0** (the older v2.2 retires Aug 3, 2026 — use v3.0). It's a *dynamic PDF*, which trips up everyone the first time:

1. **Don't click the link to open it in a browser — it won't work.** On macOS Safari: **Control-click → Download Linked File**. On Windows: right-click → "Save Link As." Save it locally first.
2. **Open it in Adobe Acrobat Pro** (recommended) or PDF-XChange Editor (≥10.2) or Foxit (≥12.1). Plain Adobe Reader will open it but **can't save your edits or attachments** — so use Acrobat Pro.
3. If Acrobat runs slowly on the dynamic PDF, disable Protected Mode (Preferences → Security (Enhanced)) or use the 64-bit version / PDF-XChange.
4. Keep attachments on a **local drive** (not a network/shared drive) when adding them, to avoid corruption.

D Stage D — Fill PreSTAR and attach your package

1. Choose the application type **Q-Submission** → **Pre-Submission**. PreSTAR then shows only the relevant sections (it's adaptive).
2. Fill the form fields (sponsor, contact, product description summary, your questions). The built-in help text guides each field.
3. Attach your documents at the relevant prompts: cover letter, product/function description, intended use, multiple-function analysis, the revised regulatory rationale, model card, validation summary, change-control plan, usability note, screenshots, clinician sign-off, meeting request. Combine similar items into single PDFs where sensible; use clear file names (e.g., `1.01 Cover Letter.pdf`).
4. Mind the limits: **1 GB per attachment, 4 GB total**. Compress screenshots/video.
5. Watch the status banner at the top. It must read **"eSTAR COMPLETE."** If it says "INCOMPLETE," the Portal will reject it — fill the flagged gaps.
6. Sign the Truthful & Accurate statement last (adding/removing attachments after signing requires clearing the signature).

TIP — COLLABORATE SEQUENTIALLY, NOT SIMULTANEOUSLY

Dynamic PDFs don't support co-authoring. If an advisor is helping, pass the file one editor at a time, or print a static PDF for them to mark up and you transcribe.

E Stage E — Submit through the CDRH Portal (≈1 hour)

1. Log in to the CDRH Portal, start a new submission, choose the Q-Submission / Pre-Submission type, and upload your completed PreSTAR PDF (plus any oversized data package per Portal instructions).
2. Submit. You'll receive an **acknowledgement and a tracking number** — save these.
3. Note: *Pre-Sub meeting minutes and presentation slides* are submitted later in eCopy format via the Portal (see Stage G).

F Stage F — What FDA does next (the ~70-day clock)

Milestone	Rough timing	What happens
Acknowledgement	Days 0–7	FDA confirms receipt, assigns a submission number.
Acceptance / technical screen	~15 days	FDA checks the submission is complete and on-topic. Fix promptly if they flag something.
Lead reviewer assigned	Early	A reviewer (often with DHCoE input) takes ownership.
Written feedback	~Day 70	FDA sends written answers to your questions — usually a few days <i>before</i> the meeting. This is the deliverable you came for.
Meeting (if requested)	~Day 70–75	Teleconference to discuss the written feedback.

G Stage G — The meeting and the record

1. **Read the written feedback carefully when it lands** (days before the meeting). Prepare 3–5 clarifying points — don't relitigate, clarify.
2. **Hold the teleconference.** Keep your side small and calm. Take detailed notes.
3. **Submit your meeting minutes.** For Pre-Subs, the sponsor typically documents the meeting; submit your minutes (and any slides) via the Portal in eCopy format, generally within about 15 days. These minutes become part of your regulatory record.
4. **File everything.** The written feedback + your minutes are your evidence that FDA engaged with — and ideally concurred on — your Non-Device determination. Store them with your regulatory documents and reference them (accurately, never as "approval") with investors.

DO THIS — KEEP A ONE-PAGE SUBMISSION LOG

Track: submission number, dates of acknowledgement/acceptance/feedback/meeting, reviewer name, and links to every document version you sent. When you later raise money or talk to a partner, this log is the artifact that shows you ran a real FDA process.

Timeline & parallel work

Because FDA's clock is ~70 days, the smart play is to compress your prep, file, and let the wait run while you do everything else.

8.1 A realistic schedule

When	You	Why now
Week 1	Run Policy Navigator; email Digital Health mailbox; line up the clinician ; register CDRH Portal account.	The clinician and Portal account have the longest lead times — start both immediately.
Weeks 1–2	Fix the brief (Part 3); finalize the four core package docs (drafts provided).	Bulk of the writing; mostly reuse + reconcile.
Week 2	Wire the CI guard; capture screenshots; write usability + validation + change-control notes.	Engineering + evidence, parallelizable.
Week 3	Clinician signs off; optional flat-fee review of intended use + questions; assemble PreSTAR.	Reviews converge; package comes together.
End of Week 3	Submit.	Clock starts.
Weeks 4–13	Investor outreach, sales, product — <i>in parallel</i> .	FDA is reviewing; you lose nothing by waiting productively.
~Week 13–14	Receive written feedback; hold meeting; file minutes.	The payoff.

8.2 Decision gates

- **Gate 1 (before filing):** Has a human with regulatory sense read the intended use and questions? If no, pause and get that read — it's the cheapest insurance you'll buy.
- **Gate 2 (before filing):** Is there at least one signed clinician entry? If no, you're not ready.
- **Gate 3 (before filing):** Does every cited exhibit actually exist and attach? If no, fix references.

Red flags that flip you to "device"

Keep this list taped to your monitor. Any one of these can turn a Non-Device function into a regulated device — or sink the Pre-Sub's credibility.

If ARKA-CLIN does this...	...the consequence
Analyzes image pixels or physiological signals/waveforms	Fails Criterion 1 → device. Keep the viewer firewalled and the CI guard live.
Outputs a single directive the clinician is meant to follow without review	At best enforcement discretion (weaker); at worst a device. Stay multi-option for statutory exclusion.
Pushes time-critical alerts (sepsis, stroke, STEMI)	Fails Criterion 4 (2026 placement) → device. Stay elective/non-urgent.
Hides its basis / can't be independently reviewed	Fails Criterion 4 → device. Keep citations + SHAP visible and usable.
Goes patient/consumer-facing	Breaks the HCP-facing framing → likely device. Keep CLIN behind clinician auth.
Auto-places or blocks orders	Converts recommendation to directive → device-like. Stay non-blocking.
Markets as "FDA-approved/cleared"	Misbranding regardless of classification. Never do this.
Relies on novel/proprietary "evidence" instead of accepted sources	Weakens Criterion 2 under the 2026 emphasis. Cite accepted guidelines/literature.

WHY THIS LIST IS YOUR FRIEND

Every item is something you can *control by design*. The reason ARKA has a credible Non-Device story is that your architecture already avoids most of these. The Pre-Sub is just you showing FDA, on paper, that you've drawn these lines on purpose.

Appendices

Appendix A — Printable master checklist

Done	Item
☐	Pre-work: Ran Digital Health Policy Navigator for CLIN and INS; saved screenshots
☐	Emailed Digital Health mailbox for informal steer
☐	Registered CDRH Portal account (consistent name/email)
☐	Brief fixes: Renamed AIIE → ARKA-CLIN/ARKA-INS everywhere
☐	Reframed INS as administrative-support (A); removed INS from CDS argument
☐	Reconciled intended use (adult vs adult+pediatric) across all docs
☐	Fixed guidance citation (Jan 6/29 2026; dropped GUI01400062)
☐	Reconciled "never blocks" vs INS critical/override language
☐	Re-worded validation from "complete" to "synthetic; real-world planned"
☐	Made evidence-source policy (ACR vs non-ACR) consistent
☐	Shifted tone from "verdict" to "request for concurrence"
☐	One consistent professional contact across all docs + Portal
☐	Package: Cover letter
☐	Product/function description (with data-flow diagram + annotated card)
☐	Intended-use / indications statement (human-reviewed)
☐	3–4 questions (concurrence-style, ≤4 topics)
☐	Multiple-function analysis (CLIN/INS/viewer map)
☐	Meeting request (type, format, 3+ dates, agenda w/ times, attendees, non-US flagged)
☐	Evidence: Revised regulatory rationale; model card; data-input inventory; screenshots
☐	Gap-fillers: ≥1 signed clinician entry
☐	INS administrative-support memo
☐	Usability/human-factors note
☐	Honest validation summary
☐	Change-control plan (rules/catalogue/weights)
☐	Criterion-1 CI guard wired + passing screenshot

☐	Labeling/no-misbranding attestation
☐	File: PreSTAR2 v3.0 says "eSTAR COMPLETE"; submitted; tracking # saved

Appendix B — The four Non-Device CDS criteria (ARKA map + 2022 → 2026 change)

Crit.	Statutory test	What changed in 2026	ARKA-CLIN posture
1	Not intended to acquire/process/analyze an image, IVD signal, or acquisition pattern.	Unchanged interpretation.	Structured data only; viewer fenced; CI guard.
2	Displays/analyzes medical info (incl. peer-reviewed studies, guidelines).	Stronger emphasis on "well-understood and accepted sources."	Citation on every card; consistent sources.
3	Supports/provides recommendations to an HCP (not a directive).	Single-output now gets enforcement discretion; time-critical limit <i>moved out</i> of C3.	Multi-option → statutory exclusion (stronger lane).
4	Enables HCP to independently review the basis; not to rely primarily on it.	Stronger <i>usability</i> emphasis; time-critical concern now lives <i>here</i> .	Tiered transparency; explicitly non-time-critical.

Appendix C — §520(o)(1) quick reference

¶	Excluded software (not a device)
(A)	Administrative support of a health care facility (billing/claims, eligibility determination, scheduling, business/population-health analytics, claims-based utilization/cost analysis). → ARKA-INS
(B)	Maintaining/encouraging a healthy lifestyle, unrelated to diagnosis/treatment.
(C)	Serving as electronic patient records.
(D)	Transferring, storing, converting formats, or displaying data/results.
(E)	Clinical decision support meeting all four criteria. → ARKA-CLIN

Appendix D — Glossary

Term	Meaning
510(k)	A premarket notification to market a device. You're avoiding this by being non-device.
513(g)	A formal (fee-bearing) request for FDA's view on classification. The paid alternative to a free Pre-Sub.
CDRH	FDA's Center for Devices and Radiological Health — handles devices and Q-Subs.

DHCoE	Digital Health Center of Excellence — FDA's digital-health group; runs the Digital Health mailbox.
DICE	Division of Industry & Consumer Education — CDRH's free help desk.
eSTAR / PreSTAR	FDA's interactive PDF templates; PreSTAR is the one for Pre-Subs/Q-Subs (current: PreSTAR2 v3.0).
Enforcement discretion	FDA's choice not to enforce device rules on something that technically is a device — weaker than exclusion.
Q-Sub	Q-Submission — the umbrella program; a Pre-Sub is the most common type.
SaMD	Software as a Medical Device — the broad category Non-Device CDS sits adjacent to.
Statutory exclusion	Not a device by law (the strong lane ARKA-CLIN targets).

Appendix E — Key links

Resource	URL
CDS Software Guidance (Jan 2026)	fda.gov/media/109618/download
Q-Submission Program Guidance (May 2025)	fda.gov/regulatory-information/search-fda-guidance-documents/requests-feedback-and-meetings-medical-device-submissions-q-submission-program
eSTAR / PreSTAR program (download template)	fda.gov/medical-devices/how-study-and-market-your-device/estar-program
CDRH Portal (register & submit)	fda.gov/medical-devices/industry-medical-devices/send-and-track-medical-device-premarket-submissions-online-cdrh-portal
Digital Health Policy Navigator	fda.gov/medical-devices/digital-health-center-excellence/digital-health-policy-navigator
CDS Software FAQs	fda.gov/medical-devices/software-medical-device-samd/clinical-decision-support-software-frequently-asked-questions-faqs
Multiple Function Device Products guidance	fda.gov/media/112671/download
Digital Health mailbox	DigitalHealth@fda.hhs.gov
Q-Sub / submission support	OPEQSubmissionSupport@fda.hhs.gov DICE: 1-800-638-2041

Appendix F — Copy-paste email templates

F1 — Informal note to the Digital Health mailbox

Subject: Non-Device CDS framing question — ARKA-CLIN (imaging appropriateness)

Hello — I'm the founder of ARKA Health, a small digital-health startup. We've built a clinician-facing imaging-appropriateness decision support tool (ARKA-CLIN) that uses only structured data, cites

published guidelines, presents ranked options with the basis shown, and leaves the decision to the clinician. We believe it meets the four Non-Device CDS criteria under §520(o)(1)(E). A separate module (ARKA-INS) performs benefit-eligibility and claims-based functions we consider administrative support under §520(o)(1)(A). Before filing a Pre-Submission, is there any informal feedback you can share on this framing or on what you'd want to see in our questions? Thank you for your time.

F2 — Request to a clinician for sign-off

Subject: ~2–3 hours of clinical review for an FDA submission (paid/advisory)

Hi Dr. [Name] — I'm preparing an FDA Pre-Submission for a clinician-facing imaging-appropriateness tool. I need a licensed clinician to review our rule logic, the guideline citations behind each rule, our feature list, and our card wording, and to sign a short dated attestation (name, credentials, license #). It's a focused, time-boxed review — I'll send a packet and a checklist. Would you be open to this on an advisory or hourly basis?

F3 — Scoping a flat-fee consultant review

Subject: Fixed-fee review of two pages (digital-health Pre-Sub)

*Hi [Name] — I'm filing an FDA Pre-Submission myself and want one expert read of just two artifacts: our intended-use statement and our four FDA questions (~2 pages total). I'm looking for a **fixed price** to review and redline these, returned within a week — not an ongoing engagement. Could you quote that scope?*

Appendix G — Sources & further reading

FDA, *Clinical Decision Support Software* — final guidance issued Jan 6, 2026 (re-issued Jan 29, 2026), [fda.gov/media/109618/download](https://www.fda.gov/media/109618/download). • FDA, *Requests for Feedback and Meetings for Medical Device Submissions: The Q-Submission Program* (final, May 29, 2025). • FDA, *eSTAR Program* page (updated June 1, 2026; PreSTAR2 v3.0). • FDA, *CDS Software FAQs & Digital Health Policy Navigator*. • FDA, *Multiple Function Device Products: Policy and Considerations*. • FD&C Act §520(o)(1)(A)–(E) (21st Century Cures Act §3060). • Covington & Burling, "5 Key Takeaways from FDA's Revised CDS Software Guidance" (Jan 8, 2026). • Your repository: `REGULATORY_RATIONALE_MEMO.md`, `REGULATORY_BASELINE_AUDIT.md`, `SCOPE_BOUNDARY.md`, `CLINICAL_SIGN_OFF_LOG.md`, `ml-service/MODEL_CARD.md`, `INS_ARCHITECTURE.md`, `CDS_SANDBOX_REGISTRATION.md`.

Disclaimer. This playbook is an informational field guide prepared to help you navigate FDA's Pre-Submission process. It is not legal or regulatory advice, and it does not create any attorney/consultant relationship. FDA's determinations depend on your specific intended use and evidence. Have a qualified human review your package before filing, and rely on FDA's official guidance documents as the controlling source.

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